

Pre Vs. Post Refinement Comparison:

Validation parameter	Before	After	Change
ERRAT quality factor	80.8728	97.7444	+16.8716 (significant improvement)
Favoured regions (%)	82.5	95.2	+12.7 (improved)
Allowed regions (%)	16.6	4.8	-11.8 (improved)
Outliers (%)	0	0	0 (no change)
Verify 3D result	93.64%	87.94%	-5.7% (slight decrease)

A higher ERRAT score indicates a better-quality model, with values closer to 100 suggesting improved structural reliability. In this case, the increase from 80.87 to 97.74 indicates a substantial improvement after refinement.

The Ramachandran Favoured Regions (%) represent the percentage of amino acid residues whose backbone dihedral angles fall within the most energetically favourable regions of the Ramachandran plot. A higher percentage indicates better stereochemical quality. The increase from 82.5% to 95.2% therefore indicates improved backbone geometry.

The Allowed Regions (%) represent residues whose backbone angles fall within regions that are sterically acceptable but less energetically favourable than the favoured regions. A decrease is generally desirable because it indicates that more residues have moved into the optimal regions. The decrease from 16.6% to 4.8% therefore represents an improvement.

Outliers (%) indicate the percentage of residues whose dihedral angles fall outside the generally allowed regions of the Ramachandran plot. These residues may have unusual or unfavorable backbone conformations. Ideally, this value should be as close to 0% as possible. Since the value remained 0% before and after refinement, there was no change and the model maintained good stereochemical quality in this respect.

The Verify3D score assesses how well each amino acid's local three-dimensional environment agrees with the expected environment for that amino acid based on known protein structures. A higher percentage indicates better compatibility between the amino acid sequence and its three-dimensional structural environment. The decrease from 93.64% to 87.94% represents a slight deterioration in this parameter, although the post-refinement value still indicates generally good sequence–structure compatibility.

Overall, the validation results suggest that GalaxyRefine substantially improved the structural quality of chain A, particularly in terms of backbone stereochemistry. Although Verify3D showed a modest decrease, the strong improvements in ERRAT and Ramachandran statistics, together with the maintenance of 0% outliers, indicate that the refined model is overall better than the initial model.